INCH-POUND

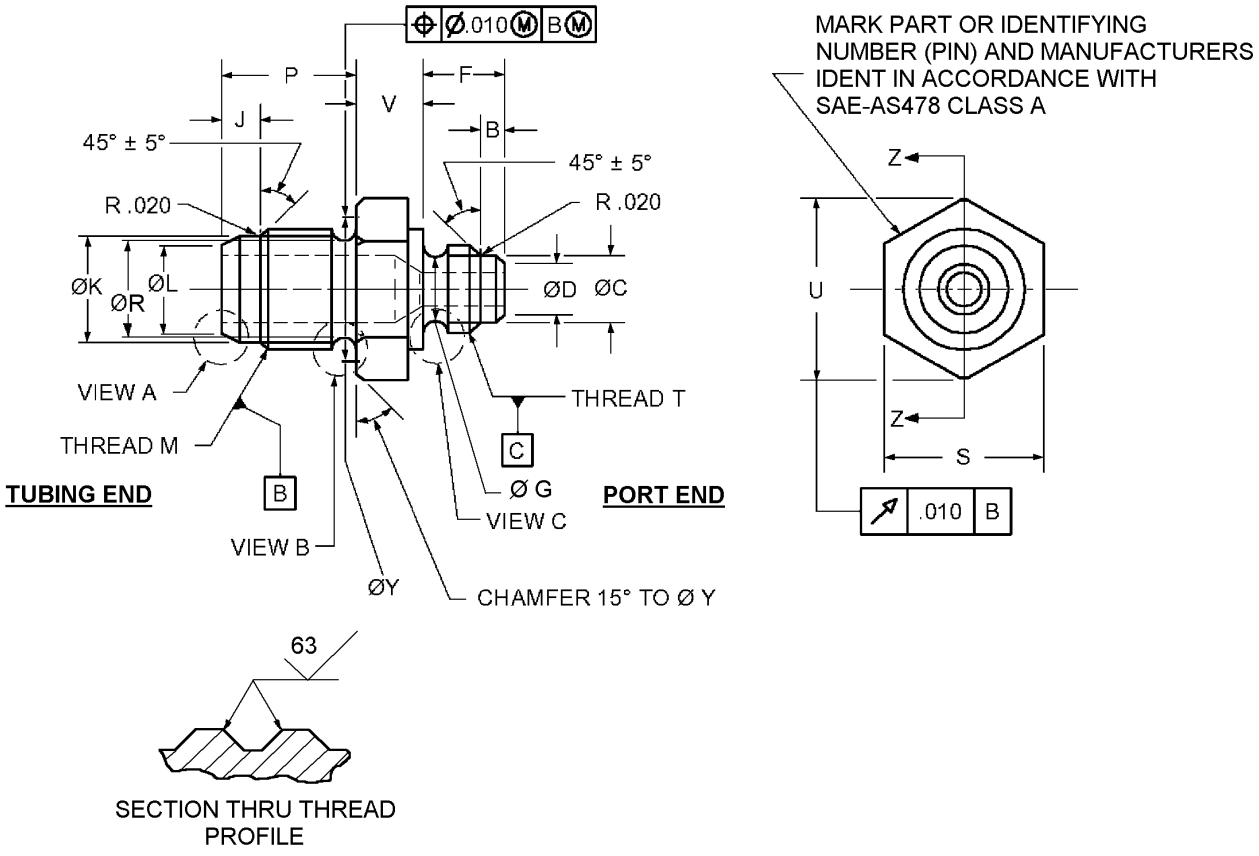
MS9769B
w/AMENDMENT 2
4 May 2017
SUPERSEDING
MS9769B
w/AMENDMENT 1
12 October 2011

DETAIL SPECIFICATION SHEET

REDUCER, TUBE - STRAIGHT, TUBE TO BOSS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



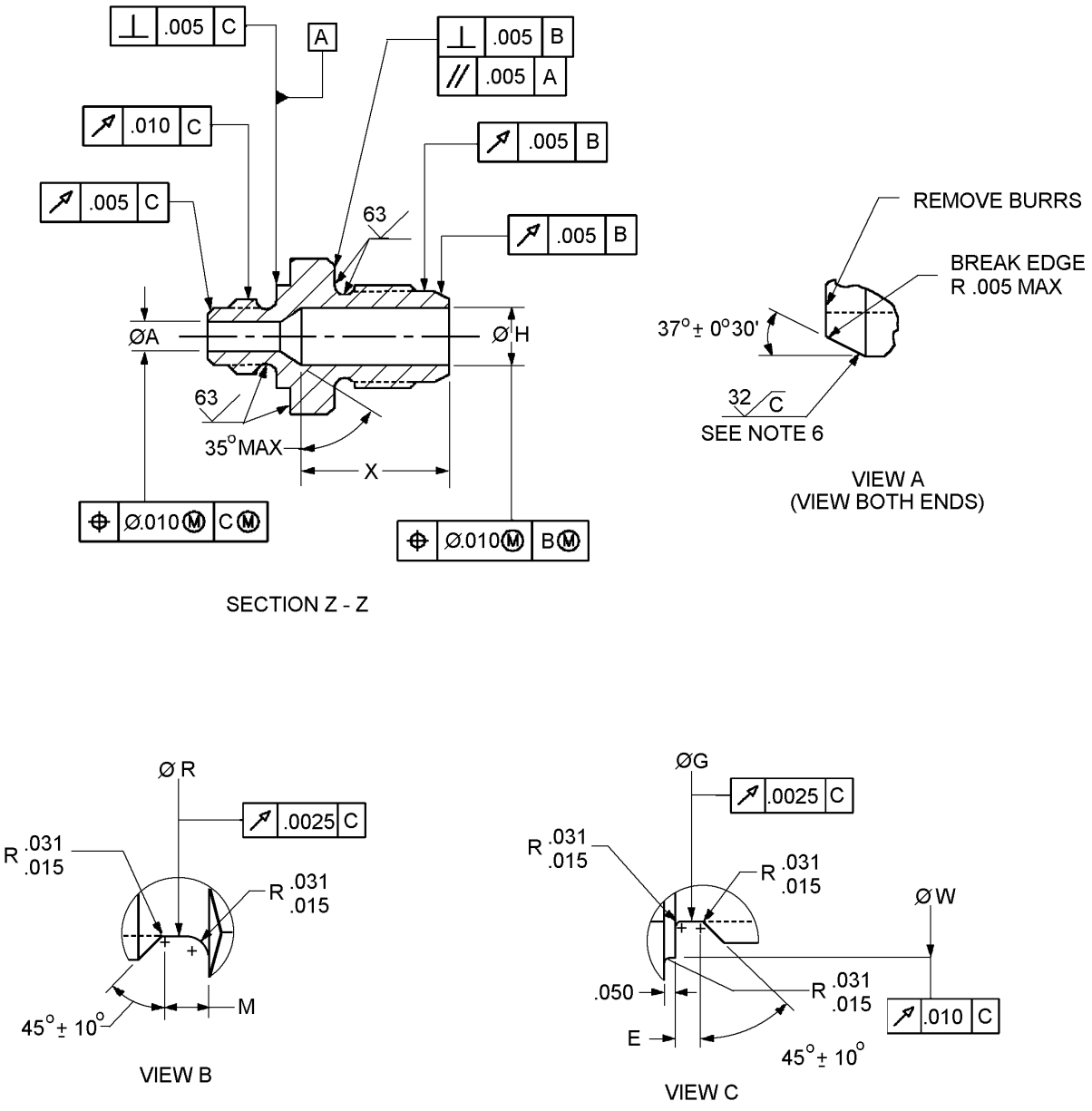
See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations.

AMSC N/A

FSC 4730

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Inches	mm
.0025	0.064
.005	0.13
.010	0.25
.015	0.38
.020	0.51
.031	0.79
.050	1.27

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:24Z
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PIN	Nom tube size inches (mm)	Port size inches (mm)	Thread M SAE-AS8879	Thread T SAE-AS8879
MS9769-021	.125 (3.18)	.188 (4.78)	.375-24UNJF-3A	.3125-24UNJF-3A
MS9769-022	.125 (3.18)	.250 (6.35)	.4375-20UNJF-3A	.3125-24UNJF-3A
MS9769-023	.125 (3.18)	.375 (9.53)	.5625-18UNJF-3A	.3125-24UNJF-3A
MS9769-024	.125 (3.18)	.500 (12.70)	.750-16UNJF-3A	.3125-24UNJF-3A
MS9769-031	.188 (4.78)	.250 (6.35)	.4375-20UNJF-3A	.375-24UNJF-3A
MS9769-032	.188 (4.78)	.312 (7.92)	.500-20UNJF-3A	.375-24UNJF-3A
MS9769-033	.188 (4.78)	.375 (9.53)	. 5625-18UNJF-3A	.375-24UNJF-3A
MS9769-034	.188 (4.78)	.500 (12.70)	.750-16UNJF-3A	.375-24UNJF-3A
MS9769-035	.188 (4.78)	.625 (15.88)	.875-14UNJF-3A	.375-24UNJF-3A
MS9769-036	.188 (4.78)	.750 (19.05)	1.0625-12UNJ-3A	.375-24UNJF-3A
MS9769-037	.188 (4.78)	1.000 (25.40)	1.3125-12UNJ-3A	.375-24UNJF-3A
MS9769-041	.250 (6.35)	.312 (7.92)	.500-20UNJF-3A	.4375-20UNJF-3A
MS9769-042	.250 (6.35)	.375 (9.53)	.5625-18UNJF-3A	.4375-20UNJF-3A
MS9769-043	.250 (6.35)	.438 (11.13)	.625-18UNJF-3A	.4375-20UNJF-3A
MS9769-044	.250 (6.35)	.500 (12.70)	.750-16UNJF-3A	.4375-20UNJF-3A
MS9769-045	.250 (6.35)	.625 (15.88)	.875-14UNJF-3A	.4375-20UNJF-3A
MS9769-046	.250 (6.35)	.750 (19.05)	1.0625-12UNJ-3A	.4375-20UNJF-3A
MS9769-051	.312 (7.92)	.375 (9.53)	.5625-18UNJF-3A	.500-20UNJF-3A
MS9769-052	.312 (7.92)	.438 (11.13)	.625-18UNJF-3A	.500-20UNJF-3A
MS9769-061	.375 (9.53)	.438 (11.13)	.625-18UNJF-3A	.5625-18UNJF-3A
MS9769-062	.375 (9.53)	.500 (12.70)	.750-16UNJF-3A	.5625-18UNJF-3A
MS9769-063	.375 (9.53)	.562 (14.27)	.8125-16UNJ-3A	.5625-18UNJF-3A
MS9769-064	.375 (9.53)	.625 (15.88)	.875-14UNJF-3A	.5625-18UNJF-3A
MS9769-065	.375 (9.53)	.750 (19.05)	1.0625-12UNJ-3A	.5625-18UNJF-3A
MS9769-071	.438 (11.13)	.500 (12.70)	.750-16UNJF-3A	.625-18UNJF-3A
MS9769-081	.500 (12.70)	.562 (14.27)	.8125-16UNJ-3A	.750-16UNJF-3A
MS9769-082	.500 (12.70)	.625 (15.88)	.875-14UNJF-3A	.750-16UNJF-3A
MS9769-083	.500 (12.70)	.688 (17.48)	1.000-12UNJ-3A	.750-16UNJF-3A
MS9769-084	.500 (12.70)	.750 (19.05)	1.0625-12UNJ-3A	.750-16UNJF-3A
MS9769-085	.500 (12.70)	1.000 (25.40)	1.3125-12UNJ-3A	.750-16UNJF-3A
MS9769-121	.750 (19.05)	1.000 (25.40)	1.3125-12UNJ-3A	1.0625-12UNJF-3A
MS9769-122	.750 (19.05)	1.250 (31.75)	1.625-12UNJ-3A	1.0625-12UNJF-3A
MS9769-161	1.000 (25.40)	1.250 (31.75)	1.625-12UNJ-3A	1.3125-12UNJF-3A

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:24Z
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PIN	A dia. inches (mm)	B +.015 (0.38) -.000 inches (mm)	C dia. +.000 -.010 (0.25) inches (mm)	D dia. ±.003 (0.08) inches (mm)
MS9769-021	.058 - .065 (1.47 - 1.65)	.144 (3.66)	.245 (6.22)	.132 (3.35)
MS9769-022	.058 - .065 (1.47 - 1.65)	.144 (3.66)	.245 (6.22)	.132 (3.35)
MS9769-023	.058 - .065 (1.47 - 1.65)	.144 (3.66)	.245 (6.22)	.132 (3.35)
MS9769-024	.058 - .065 (1.47 - 1.65)	.144 (3.66)	.245 (6.22)	.132 (3.35)
MS9769-031	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-032	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-033	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-034	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-035	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-036	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-037	.121 - .128 (3.07 - 3.25)	.144 (3.66)	.307 (7.80)	.195 (4.95)
MS9769-041	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-042	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-043	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-044	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-045	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-046	.168 - .175 (4.27 - 4.45)	.160 (4.06)	.359 (9.12)	.242 (6.15)
MS9769-051	.230 - .237 (5.84 - 6.02)	.160 (4.06)	.421 (10.69)	.304 (7.72)
MS9769-052	.230 - .237 (5.84 - 6.02)	.160 (4.06)	.421 (10.69)	.304 (7.72)
MS9769-061	.293 - .301 (7.44 - 7.65)	.165 (4.19)	.476 (12.09)	.368 (9.35)
MS9769-062	.293 - .301 (7.44 - 7.65)	.165 (4.19)	.476 (12.09)	.368 (9.35)
MS9769-063	.293 - .301 (7.44 - 7.65)	.165 (4.19)	.476 (12.09)	.368 (9.35)
MS9769-064	.293 - .301 (7.44 - 7.65)	.165 (4.19)	.476 (12.09)	.368 (9.35)
MS9769-065	.293 - .301 (7.44 - 7.65)	.165 (4.19)	.476 (12.09)	.368 (9.35)
MS9769-071	.355 - .363 (9.02 - 9.22)	.165 (4.19)	.539 (13.69)	.430 (10.92)
MS9769-081	.387 - .395 (9.83 - 10.03)	.229 (5.82)	.654 (16.61)	.462 (11.73)
MS9769-082	.387 - .395 (9.83 - 10.03)	.229 (5.82)	.654 (16.61)	.462 (11.73)
MS9769-083	.387 - .395 (9.83 - 10.03)	.229 (5.82)	.654 (16.61)	.462 (11.73)
MS9769-084	.387 - .395 (9.83 - 10.03)	.229 (5.82)	.654 (16.61)	.462 (11.73)
MS9769-085	.387 - .395 (9.83 - 10.03)	.229 (5.82)	.654 (16.61)	.462 (11.73)
MS9769-121	.604 - .614 (15.34 - 15.60)	.304 (7.72)	.938 (23.83)	.681 (17.30)
MS9769-122	.604 - .614 (15.34 - 15.60)	.304 (7.72)	.938 (23.83)	.681 (17.30)
MS9769-161	.839 - .851 (21.31 - 21.62)	.312 (7.92)	1.188 (30.18)	.918 (23.32)

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:24Z
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PIN	E +.015 (0.38) -.000 inches (mm)	F inches (mm)	G dia. +.002 (0.05) -.003 (0.08) inches (mm)	H dia. inches (mm)
MS9769-021	.063 (1.60)	.415 (10.54)	.250 (6.35)	.121 - .128 (3.07 - 3.25)
MS9769-022	.063 (1.60)	.415 (10.54)	.250 (6.35)	.168 - .175 (4.27 - 4.45)
MS9769-023	.063 (1.60)	.415 (10.54)	.250 (6.35)	.293 - .301 (7.44 - 7.65)
MS9769-024	.063 (1.60)	.415 (10.54)	.250 (6.35)	.387 - .395 (9.83 - 10.03)
MS9769-031	.063 (1.60)	.446 (11.33)	.312 (7.92)	.168 - .175 (4.27 - 4.45)
MS9769-032	.063 (1.60)	.446 (11.33)	.312 (7.92)	.230 - .237 (5.84 - 6.02)
MS9769-033	.063 (1.60)	.446 (11.33)	.312 (7.92)	.293 - .301 (7.44 - 7.65)
MS9769-034	.063 (1.60)	.446 (11.33)	.312 (7.92)	.387 - .395 (9.83 - 10.03)
MS9769-035	.063 (1.60)	.446 (11.33)	.312 (7.92)	.480 - .488 (12.19 - 12.40)
MS9769-036	.063 (1.60)	.446 (11.33)	.312 (7.92)	.604 - .614 (15.34 - 15.60)
MS9769-037	.063 (1.60)	.446 (11.33)	.312 (7.92)	.839 - .851 (21.31 - 21.62)
MS9769-041	.075 (1.91)	.517 (13.13)	.364 (9.25)	.230 - .237 (5.84 - 6.02)
MS9769-042	.075 (1.91)	.517 (13.13)	.364 (9.25)	.293 - .301 (7.44 - 7.65)
MS9769-043	.075 (1.91)	.517 (13.13)	.364 (9.25)	.355 - .363 (9.02 - 9.22)
MS9769-044	.075 (1.91)	.517 (13.13)	.364 (9.25)	.387 - .395 (9.83 - 10.03)
MS9769-045	.075 (1.91)	.517 (13.13)	.364 (9.25)	.480 - .488 (12.19 - 12.40)
MS9769-046	.075 (1.91)	.517 (13.13)	.364 (9.25)	.604 - .614 (15.34 - 15.60)
MS9769-051	.075 (1.91)	.517 (13.13)	.426 (10.82)	.293 - .301 (7.44 - 7.65)
MS9769-052	.075 (1.91)	.517 (13.13)	.426 (10.82)	.355 - .363 (9.02 - 9.22)
MS9769-061	.083 (2.11)	.523 (13.28)	.481 (12.22)	.355 - .363 (9.02 - 9.22)
MS9769-062	.083 (2.11)	.523 (13.28)	.481 (12.22)	.387 - .395 (9.83 - 10.03)
MS9769-063	.083 (2.11)	.523 (13.28)	.481 (12.22)	.434 - .442 (11.02 - 11.23)
MS9769-064	.083 (2.11)	.523 (13.28)	.481 (12.22)	.480 - .488 (12.19 - 12.40)
MS9769-065	.083 (2.11)	.523 (13.28)	.481 (12.22)	.604 - .614 15.34 - 15.60)
MS9769-071	.083 (2.11)	.523 (13.28)	.544 (13.82)	.387 - .395 (9.83 - 10.03)
MS9769-081	.094 (2.39)	.633 (16.08)	.660 (16.76)	.434 - .442 (11.02 - 11.23)
MS9769-082	.094 (2.39)	.633 (16.08)	.660 (16.76)	.480 - .488 (12.19 - 12.40)
MS9769-083	.094 (2.39)	.633 (16.08)	.660 (16.76)	.542 - .552 (13.77 - 14.02)
MS9769-084	.094 (2.39)	.633 (16.08)	.660 (16.76)	.604 - .614 15.34 - 15.60)
MS9769-085	.094 (2.39)	.633 (16.08)	.660 (16.76)	.839 - .851 (21.31 - 21.62)
MS9769-121	.125 (3.18)	.853 (21.67)	.945 (24.00)	.839 - .851 (21.31 - 21.62)
MS9769-122	.125 (3.18)	.853 (21.67)	.945 (24.00)	1.073 - 1.086 (27.25 - 27.54)
MS9769-161	.125 (3.18)	.908 (23.06)	1.195 (30.35)	1.073 - 1.086 (27.25 - 27.54)

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:24Z
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PIN	J +.015 (0.38) -.000 inches (mm)	K dia. +.000 -.010 (0.25) inches (mm)	L dia. ±.003 (0.08) inches (mm)	M +.015 (0.38) -.000 inches (mm)	P
MS9769-021	.144 (3.66)	.307 (7.80)	.195 (4.95)	.063 (1.60)	.446 (11.33)
MS9769-022	.160 (4.06)	.359 (9.12)	.242 (6.15)	.075 (1.91)	.517 (13.13)
MS9769-023	.165 (4.19)	.476 (12.09)	.368 (9.35)	.083 (2.11)	.523 (13.28)
MS9769-024	.229 (5.82)	.654 (16.61)	.462 (11.73)	.094 (2.39)	.633 (16.08)
MS9769-031	.160 (4.06)	.359 (9.12)	.242 (6.15)	.075 (1.91)	.517 (13.13)
MS9769-032	.160 (4.06)	.421 (10.69)	.304 (7.72)	.075 (1.91)	.517 (13.13)
MS9769-033	.165 (4.19)	.476 (12.09)	.368 (9.35)	.083 (2.11)	.523 (13.28)
MS9769-034	.229 (5.82)	.654 (16.61)	.462 (11.73)	.094 (2.39)	.633 (16.08)
MS9769-035	.255 (6.48)	.767 (19.48)	.555 (14.10)	.107 (2.72)	.747 (18.97)
MS9769-036	.304 (7.72)	.938 (23.83)	.681 (17.30)	.125 (3.18)	.853 (21.67)
MS9769-037	.312 (7.92)	1.188 (30.18)	.918 (23.32)	.125 (3.18)	.908 (23.06)
MS9769-041	.160 (4.06)	.421 (10.69)	.304 (7.72)	.075 (1.91)	.517 (13.13)
MS9769-042	.165 (4.19)	.476 (12.09)	.368 (9.35)	.083 (2.11)	.523 (13.28)
MS9769-043	.165 (4.19)	.539 (13.69)	.430 (10.92)	.083 (2.11)	.523 (13.28)
MS9769-044	.229 (5.82)	.654 (16.61)	.462 (11.73)	.094 (2.39)	.633 (16.08)
MS9769-045	.255 (6.48)	.767 (19.48)	.555 (14.10)	.107 (2.72)	.747 (18.97)
MS9769-046	.304 (7.72)	.938 (23.83)	.681 (17.30)	.125 (3.18)	.853 (21.67)
MS9769-051	.165 (4.19)	.476 (12.09)	.368 (9.35)	.083 (2.11)	.523 (13.28)
MS9769-052	.165 (4.19)	.539 (13.69)	.430 (10.92)	.083 (2.11)	.523 (13.28)
MS9769-061	.165 (4.19)	.539 (13.69)	.430 (10.92)	.083 (2.11)	.523 (13.28)
MS9769-062	.229 (5.82)	.654 (16.61)	.462 (11.73)	.094 (2.39)	.633 (16.08)
MS9769-063	.229 (5.82)	.716 (18.19)	.509 (12.93)	.094 (2.39)	.633 (16.08)
MS9769-064	.255 (6.48)	.767 (19.48)	.555 (14.10)	.107 (2.72)	.747 (18.97)
MS9769-065	.304 (7.72)	.938 (23.83)	.681 (17.30)	.125 (3.18)	.853 (21.67)
MS9769-071	.229 (5.82)	.654 (16.61)	.462 (11.73)	.094 (2.39)	.633 (16.08)
MS9769-081	.229 (5.82)	.716 (18.19)	.509 (12.93)	.094 (2.39)	.633 (16.08)
MS9769-082	.255 (6.48)	.767 (19.48)	.555 (14.10)	.107 (2.72)	.747 (18.97)
MS9769-083	.304 (7.72)	.875 (22.23)	.619 (15.72)	.125 (3.18)	.853 (21.67)
MS9769-084	.304 (7.72)	.938 (23.83)	.681 (17.30)	.125 (3.18)	.853 (21.67)
MS9769-085	.312 (7.92)	1.188 (30.18)	.918 (23.32)	.125 (3.18)	.908 (23.06)
MS9769-121	.312 (7.92)	1.188 (30.18)	.918 (23.32)	.125 (3.18)	.908 (23.06)
MS9769-122	.363 (9.22)	1.500 (38.10)	1.513 (38.43)	.125 (3.18)	.954 (24.23)
MS9769-161	.363 (9.22)	1.500 (38.10)	1.513 (38.43)	.125 (3.18)	.954 (24.23)

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

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PIN	R dia. +.002 (0.05) -.003 (0.08) inches (mm)	S inches (mm)	U Min inches (mm)	V inches (mm)
MS9769-021	.312 (7.92)	.616 - .627 (15.65 - 15.93)	.694 (17.63)	.240 (6.10)
MS9769-022	.364 (9.25)	.679 - .690 (17.25 - 17.53)	.766 (19.46)	.270 (6.86)
MS9769-023	.481 (12.22)	.803 - .814 (20.40-20.68)	.907 (23.04)	.300 (7.62)
MS9769-024	.660 (16.76)	.990 - 1.002 (15.15 - 25.45)	1.121 (28.47)	.300 (7.62)
MS9769-031	.364 (9.25)	.679 - .690 (17.25 - 17.53)	.766 (19.46)	.270 (6.86)
MS9769-032	.426 (10.82)	.741 - .752 (18.82 - 19.10)	.837 (21.26)	.270 (6.86)
MS9769-033	.481 (12.22)	.803 - .814 (20.40-20.68)	.907 (23.04)	.300 (7.62)
MS9769-034	.660 (16.76)	.990 - 1.002 (25.15 - 25.45)	1.121 (28.47)	.300 (7.62)
MS9769-035	.773 (19.63)	1.115 - 1.127 (28.32 - 28.63)	1.262 (32.05)	.360 (9.14)
MS9769-036	.945 (24.00)	1.301 - 1.314 (33.05 - 33.38)	1.475 (37.47)	.360 (9.14)
MS9769-037	1.195 (30.35)	1.614 - 1.627 (41.00 - 41.33)	1.833 (46.56)	.410 (10.41)
MS9769-041	.426 (10.82)	.741 - .752 (18.82 - 19.10)	.837 (21.26)	.270 (6.86)
MS9769-042	.481 (12.22)	.803 - .814 (20.40-20.68)	.907 (23.04)	.300 (7.62)
MS9769-043	.544 (13.82)	.865 - .877 (21.97 - 22.28)	.978 (24.84)	.300 (7.62)
MS9769-044	.660 (16.76)	.990 - 1.002 (25.15 - 25.45)	1.121 (28.47)	.300 (7.62)
MS9769-045	.773 (19.63)	1.115 - 1.127 (28.32 - 28.63)	1.262 (32.05)	.360 (9.14)
MS9769-046	.945 (24.00)	1.301 - 1.314 (33.05 - 33.38)	1.475 (37.47)	.360 (9.14)
MS9769-051	.481 (12.22)	.803 - .814 (20.40-20.68)	.907 (23.04)	.300 (7.62)
MS9769-052	.544 (13.82)	.865 - .877 (21.97 - 22.28)	.978 (24.84)	.300 (7.62)
MS9769-061	.544 (13.82)	.865 - .877 (21.97 - 22.28)	.978 (24.84)	.300 (7.62)
MS9769-062	.660 (16.76)	.990 - 1.002 (25.15 - 25.45)	1.121 (28.47)	.300 (7.62)
MS9769-063	.722 (18.34)	1.052 - 1.064 (26.72 - 27.03)	1.191 (30.25)	.300 (7.62)
MS9769-064	.773 (19.63)	1.115 - 1.127 (28.32 - 28.63)	1.262 (32.05)	.360 (9.14)
MS9769-065	.945 (24.00)	1.301 - 1.314 (33.05 - 33.38)	1.475 (37.47)	.360 (9.14)
MS9769-071	.660 (16.76)	.990 - 1.002 (25.15 - 25.45)	1.121 (28.47)	.300 (7.62)
MS9769-081	.722 (18.34)	1.052 - 1.064 (26.72 - 27.03)	1.191 (30.25)	.300 (7.62)
MS9769-082	.773 (19.63)	1.115 - 1.127 (28.32 - 28.63)	1.262 (32.05)	.360 (9.14)
MS9769-083	.882 (22.40)	1.301 - 1.314 (33.05 - 33.38)	1.475 (37.47)	.360 (9.14)
MS9769-084	.945 (24.00)	1.301 - 1.314 (33.05 - 33.38)	1.475 (37.47)	.360 (9.14)
MS9769-085	1.195 (30.35)	1.614 - 1.627 (41.00 - 41.33)	1.833 (46.56)	.410 (10.41)
MS9769-121	1.195 (30.35)	1.614 - 1.627 (41.00 - 41.33)	1.833 (46.56)	.410 (10.41)
MS9769-122	1.507 (38.28)	1.863 - 1.877 (47.32 - 47.68)	2.117 (53.77)	.410 (10.41)
MS9769-161	1.507 (38.28)	1.863 - 1.877 (47.32 - 47.68)	2.112 (53.64)	.410 (10.41)

See notes at end of figure.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

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PIN	W dia. inches (mm)	X inches (mm)	Y dia. inches (mm)	Approx weight lb/100 (kg/100)
MS9769-021	.565 (14.35)	.585 (14.86)	.625 (15.88)	5.05 (2.3)
MS9769-022	.565 (14.35)	.670 (17.02)	.688 (17.48)	6.31 (2.9)
MS9769-023	.565 (14.35)	.665 (16.89)	.812 (20.62)	7.70 (3.5)
MS9769-024	.565 (14.35)	.740 (18.80)	1.000 (25.40)	11.14 (5.1)
MS9769-031	.625 (15.88)	.690 (17.53)	.688 (17.48)	6.94 (3.1)
MS9769-032	.625 (15.88)	.670 (17.02)	.750 (19.05)	7.53 (3.4)
MS9769-033	.625 (15.88)	.685 (17.40)	.812 (20.62)	8.40 (3.8)
MS9769-034	.625 (15.88)	.765 (19.43)	1.000 (25.40)	11.87 (5.4)
MS9769-035	.625 (15.88)	.895 (22.73)	1.125 (28.58)	16.91 (7.7)
MS9769-036	.625 (15.88)	.965 (24.51)	1.312 (33.32)	20.42 (9.3)
MS9769-037	.625 (15.88)	.985 (25.02)	1.625 (41.28)	29.52 (13.4)
MS9769-041	.690 (17.53)	.690 (17.53)	.750 (19.05)	8.29 (3.8)
MS9769-042	.690 (17.53)	.710 (18.03)	.812 (20.62)	9.16 (4.2)
MS9769-043	.690 (17.53)	.705 (17.91)	.875 (22.23)	9.78 (4.4)
MS9769-044	.690 (17.53)	.770 (19.56)	1.000 (25.40)	12.57 (5.7)
MS9769-045	.690 (17.53)	.910 (23.11)	1.125 (28.58)	17.64 (8.0)
MS9769-046	.690 (17.53)	.985 (25.02)	1.312 (33.32)	21.18 (9.6)
MS9769-051	.750 (19.05)	.725 (18.42)	.812 (20.62)	9.49 (4.3)
MS9769-052	.750 (19.05)	.700 (17.78)	.875 (22.23)	10.09 (4.6)
MS9769-061	.815 (20.70)	.725 (18.42)	.875 (22.23)	10.26 (4.7)
MS9769-062	.815 (20.70)	.825 (20.96)	1.000 (25.40)	12.97 (5.9)
MS9769-063	.815 (20.70)	.808 (20.52)	1.062 (26.97)	13.80 (6.3)
MS9769-064	.815 (20.70)	.955 (24.26)	1.125 (28.58)	17.98 (8.2)
MS9769-065	.815 (20.70)	1.028 (26.11)	1.312 (33.32)	21.43 (9.7)
MS9769-071	.875 (22.23)	.845 (21.46)	1.000 (25.40)	13.12 (6.0)
MS9769-081	1.000 (25.40)	.818 (20.78)	1.062 (26.97)	15.91 (7.2)
MS9769-082	1.000 (25.40)	.965 (24.51)	1.125 (28.58)	20.09 (9.1)
MS9769-083	1.000 (25.40)	1.060 (26.92)	1.312 (33.32)	23.84 (10.8)
MS9769-084	1.000 (25.40)	1.035 (26.29)	1.312 (33.32)	23.57 (10.7)
MS9769-085	1.000 (25.40)	1.060 (26.92)	1.625 (41.28)	32.31 (14.7)
MS9769-121	1.375 (34.93)	1.120 (28.45)	1.625 (41.28)	35.93 (16.3)
MS9769-122	1.375 (34.93)	1.085 (27.56)	1.875 (47.63)	43.74 (19.8)
MS9769-161	1.625 (41.28)	1.163 (29.54)	1.875 (47.63)	42.27 (19.2)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances for linear dimensions are ±.010 inch (0.25 mm), angular dimensions ±5°.
4. Machined surfaces shall be finished to 125µ in Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
5. Break sharp edges .003 to .015 inch (0.08 to 0.38 mm), unless otherwise specified.
6. Sealing surface shall be free of flaws, chatter marks, or low spots.

FIGURE 1. Reducer, tube dimensions and configurations - Continued.

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REQUIREMENTS:

Reducer, tube shall be as specified on figure 1.

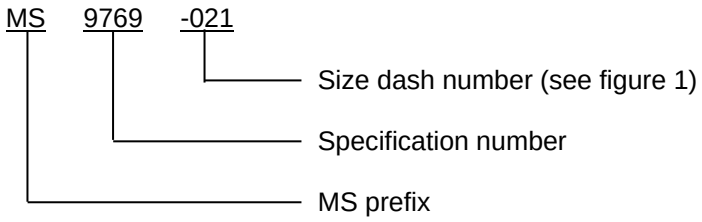
Fluorescent penetrant examination shall be in accordance with ASTM E1417/E1417M.

Fluorescent penetrant shall be in accordance with SAE-AMS2644.

Material: Corrosion resistant steel in accordance with SAE-AMS5646.

Finish: Passivate in accordance with SAE-AMS2700, type 6.

PIN: The PIN consists of the letter “MS” the specification sheet number, and a dash number for tube and port size. Unassigned PIN’s shall not be used.



Marking: Part shall be permanently marked with the MS PIN, and include the manufacturers CAGE, name, or trademark.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Application note: Fluid flow reduction is from dimension “Dia. L” to “Dia. A”.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. This document references the following:

ASME B46.1
ASTM E1417/E1417M
SAE-AMS2644
SAE-AMS2700
SAE-AMS5646
SAE-AS478
SAE-AS8879

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CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2017-057)

Review activities:
Navy - MC, SA

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